CLUSTERING

INTRODUCTION – PART I

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INTRODUCTION

The following concepts will be introduced:

- ✓ **CLUSTER ANALYSIS**
- ✓ **PURPOSES OF CLUSTER ANALYSIS**

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VMail Message	CustServ Calls	Int'l Plan	VMail Plan	Day Charge	Eve Charge	Night Charge	Intl Charge
25	1	0	1	45.07	16.78	11.01	2.7
26	1	0	1	27.47	16.62	11.45	3.7
0	0	0	0	41.38	10.3	7.32	3.29
0	2	1	0	50.9	5.26	8.86	1.78
24	3	0	1	37.09	29.62	9.57	2.03
0	0	1	0	26.69	8.76	9.53	1.92
0	1	0	0	31.37	29.89	9.71	2.35
37	0	1	1	43.96	18.87	14.69	3.02
0	4	0	0	21.95	19.42	9.4	3.43
0	0	0	0	31.91	13.89	8.82	2.46

Your friend, the **CEO OF THE TELECOM COMPANY**, thinks that partitioning the set of customers into groups is meaningful to understand how different types of customers use their phones and useful to make informed decisions on pricing, special offers, …

You tell to your friend that **CLUSTER ANALYSIS** could be effective to solve the customers grouping problem. Indeed, **CLUSTER ANALYSIS** is used for solving many practical problems.

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In particular, you tell to your friend that **CLUSTER ANALYSIS** is applied for two main purposes:

- ✓ **UNDERSTANDING**
- ✓ **UTILITY**

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- ✓ **UNDERSTANDING:** classes, or conceptually meaningful groups of objects that share common characteristics, play an important role in how people analyze and describe the world.

  This happens in biology (hierarchical classification of all living things), information retrieval (clustering users queries), business (segmenting customers for marketing activity).

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✓ **UTILITY:** provides an abstraction from individual data objects to the clusters in which these data object reside. More precisely, the goal is to find the most representative cluster prototypes (an object that is representative of the other objects in the same cluster)

Specific sub-goals are; summarization (dimensionality reduction), compression (each cluster prototype is associated an integer) and efficiently finding nearest neighbors.

However, for your friend it is not clear **WHAT CLUSTER ANALYSIS IS**?

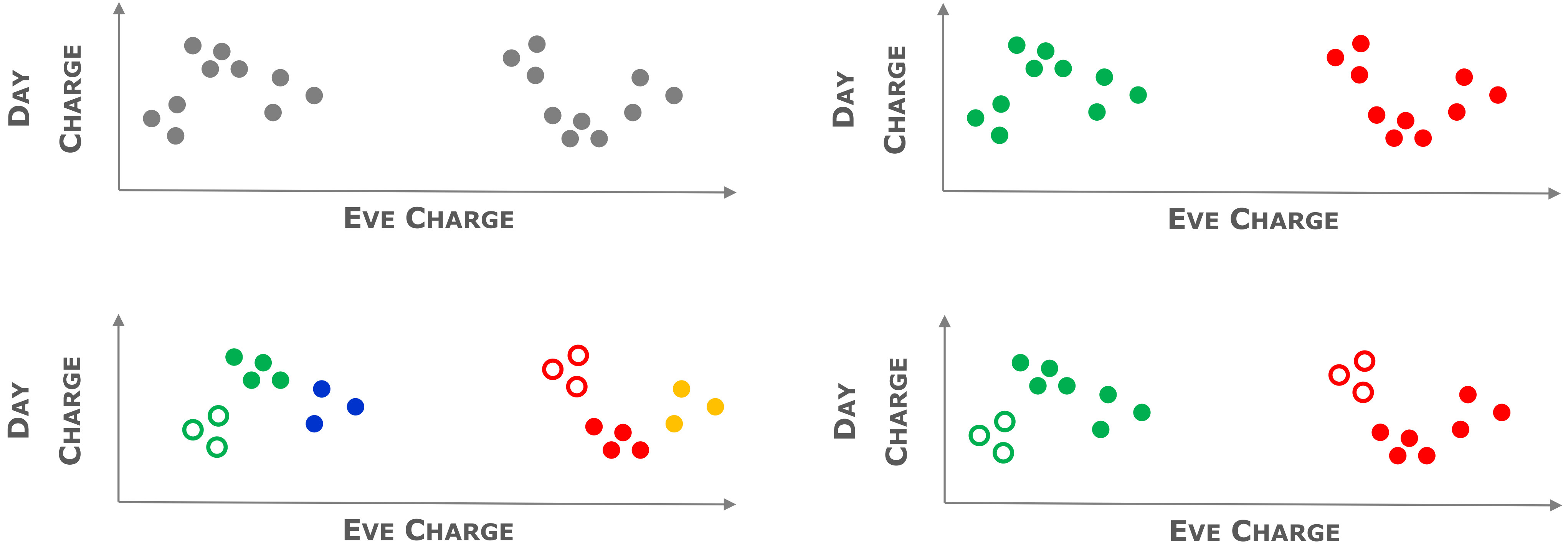
You tell your friend that **CLUSTER ANALYSIS** groups data objects based on information found in the data that describes the objects and their relationships. You explain your friend that its goal is rather simple, it is that:

- THE OBJECTS WITHIN A GROUP BE SIMILAR (RELATED) TO ONE ANOTHER AND DIFFERENT FROM (UNRELATED) THE OBJECTS IN THE OTHER GROUPS.
- THE GREATER THE SIMILARITY (OR HOMOGENEITY) WITHIN A GROUP AND THE GREATER THE DIFFERENCE BETWEEN GROUPS, THE BETTER OR MORE DISTINCT THE CLUSTERING.

WHAT DOES IT MEAN SIMILAR?

You tell your friend that in many applications the notion of cluster is not well defined.

To convince your friend of the difficulty of deciding what constitutes a cluster you use the following example:



The definition of cluster is imprecise and the best definition depends on the nature of data and the desired results.